Hamza Atiq

+92 (334) 1152668 • hamzaatiq34@gmail.com • LinkedIn • GitHub • Leetcode

Achievements

- Ranked **269/375** worldwide in the **University of California, Berkeley CALICO Spring 2026**. Link
- Participated in the **University of California, Los Angeles, CodeSprint 2026 coding competition**; team ranked **15/244** internationally. Link
- Attended **Yale Peaked Hackathon 2026**, solved **6/10** problems, and ranked **53/549** worldwide. Link
- **Harvard CS50x Puzzle Day,** solved all **9/9** puzzles (2025) and **10/10** (2026). Link
- Joined **Meta Hacker Cup 2025**, ranked **7798/13779** globally by solving complex problems in Round 1. Link
- Entered in **M(IT)^2 coding competition** and ranked **482/559** by solving a complex problem in Round 1. Link
- Participated in **10+ International Hackathons**. Certificate Links, LabLab.AI Profile, Devpost Profile
- Participated in **HackerRank Hackathon,** ranked **427/1349** globally. Link
- Solved **90+ LeetCode** algorithmic problems in Python. Link
- Developed a solar production and load prediction system using machine learning models, achieving **10/55** in the **DNN Bootcamp at GIKI.** Link
- Scored **115/160** overall (equivalent to an IELTS score of **6.5**) on the **Duolingo English Test.** Link

Education

University of Wah Website — **Wah Cantt, Pakistan**
Bachelor's in Electrical Engineering. CGPA: 3.51 — 2015-2019
Relevant Coursework: DSA, EMFT, Power Electronics, Power System Analysis, Renewable Energy Systems

Skills

- **Programming Languages:** Python
- **AI:** Generative AI, RAG systems, Agentic workflows, Prompt engineering
- **Machine Learning:** Supervised learning, model evaluation, feature engineering
- **Deep Learning frameworks:** TensorFlow, Neural networks
- **Cloud & Deployment:** AWS (S3 Bucket), FastAPI
- **Database:** PostgreSQL
- **Version Control:** Git, GitHub
- **Tools:** Visual Studio, Cursor, Google Colab
- **Interest:** Applications of Artificial Intelligence and Machine Learning.

Experience

NeuroOceans AI Website Experience — **Remote, Lahore, Pakistan**
ML Engineer — DEC 2024 – Aug 2025

- Designed and implemented a generalized web crawler to extract content-rich product data from e-commerce websites.
- Developed an AI-driven video generation system from images, created FastAPI endpoints, and integrated it with PostgreSQL for efficient data management and processing.
- Deployed ML pipelines on AWS (S3 buckets) to automate data extraction and processing, reducing manual data handling.

Family Business — **Rawalpindi, Pakistan**
Cash & Carry Retail Store | Owner — 2020-2023

- Managed daily procurement and supplier relationships for a cash and carry retail store, purchasing stock based on

real-time usage and demand tracking.
- Maintained accurate inventory and sales records in a digital system, ensuring accurate stock levels and transactions.
- Handled end-to-end store operations, including acting as a cashier during staff shortages, to keep the business running.

Vpro Engineering Website **Rawalpindi, Pakistan**
Site Engineer Aug 2019 – Feb 2020

- Installed and maintained solar energy systems on residential and commercial properties.
- Conducted site assessments to determine the feasibility of solar installations.
- Worked with a team of installers to complete projects on time and efficiently, ensuring all installations met industry safety standards and regulations.

Leadership & Activities

iCodeGuru **Remote CA, USA**
Moderator and Trainer Sep. 2025 - Present

- Moderated sessions and workshops on **IELTS** and **LeetCode**. Link
- Led 10+ **DSA** sessions voluntarily and solved **LeetCode** problems to teach students.
- Volunteered to teach **M(IT)²** past problems and guided students on how to compete effectively. Link
- Volunteered to teach **CALICO** past problems and guided students on how to compete effectively. Link

Academic Projects

GIKI SkyLab Competition: Solar Energy Production and load Forecasting | Github
Tech Stack: Python, pandas, NumPy, scikit-learn, TensorFlow, Keras, SciPy, Matplotlib, Seaborn

- Built an ML pipeline to predict solar panel generation and electrical load for 40+ solar systems across Pakistan.
- Feature engineering, including lag variables, rolling statistics, cyclical time encoding, and FFT-based frequency features to capture temporal and periodic energy patterns
- Trained and compared two models: Random Forest Regressor and a stacked LSTM with regularization.

N-1 Power Grid Contingency Screening using Graph Neural Networks | **GitHub | Results**
Tech Stack: Python, PyTorch, pandapower, pytest, Gradio, Google Colab

- Built a graph neural network in PyTorch to rank 80 possible line and transformer failures on the IEEE 57-bus test grid by risk.
- Integrated pandapower's Newton-Raphson AC power flow solver to verify high-risk cases and handle grid islanding and unsolvable contingencies.
- Evaluated the screening pipeline on 8,750 unseen test days, detecting at least 99.91% of dangerous cases in every test and achieving a median 2.25× speedup over solving all contingencies.

International Hackathons

AMD Developer Hackathon ACT III: Whisper a failing bearing motor current | GitHub | Submission Link
Tech Stack: Python, Scikit-learn, NumPy, pandas, React, TypeScript, Vercel, AMD Token Factory, Gemma-4-31B-IT

- Researched motor current bearing fault detection and the published work on the Paderborn University bearing dataset.
- Built a random forest detector that detects bearing damage in the motor current, so the motor needs no vibration

sensor.
- Deployed a web app where Gemma on AMD writes technician notes, and code checks every answer (30/30 passed, 9/9 trick questions declined).

AMD Developer Challenge: Lumi AI - Voice Helper for Elderly with Dementia | GitHub | Submission Link
Tech Stack: Python, AMD MI300X (ROCm), Qwen, QLoRA, GRPO Reinforcement Learning, ChromaDB, vLLM, HuggingFace

- Fine-tuned a Qwen-based voice companion on AMD MI300X GPUs, using QLoRA and GRPO RFL, trained on 8540 dementia specific caregiving samples.
- Built a persistent memory pipeline with ChromaDB to inject prior session context, enabling continuity of care across conversations without patient repetition.
- **Finalist** in the HuggingFace community prize track and ranked **7th/46** on the EQ-Bench.

Raise Your Hack: Personalized TutorBot | GitHub | Submission Link
Tech Stack: Python, Blackbox-powered LLMs, FastAPI, React

- Led a team of 6 members to build an AI-powered multi-agent educational assistant for students.
- Designed to make AI and programming learning accessible, engaging, and personalized.
- Integrated a Replit-based IDE that enables learners to write code and test on the same site.

Certifications

- Generative AI for Everyone. Link
- Generative AI with Diffusion Models. Link
- Neural Networks and Deep Learning. Link
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, and Optimization. Link
- Fundamentals of Deep Learning. Link